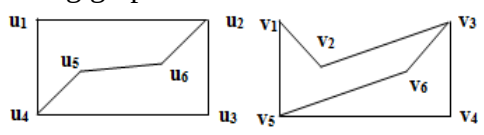
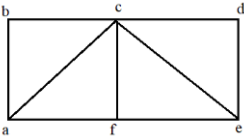
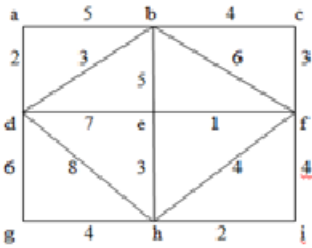


Course Code: B23BS2101					
SAGI RAMA KRISHNAM RAJU ENGINEERING COLLEGE (A)					R23
II B.Tech. I Semester MODEL QUESTION PAPER					
DISCRETE MATHEMATICS AND GRAPH THEORY					
(Common to CSE, AIML, CSBS, IT, AIDS, CSG, CIC, CSIT)					
Time: 3 Hrs.			Max. Marks: 70 M		
Answer Question No.1 compulsorily					
Answer ONE Question from EACH UNIT					
Assume suitable data if necessary					
					10 x 2 = 20 Marks
			CO	KL	M
1.	a).	Write the inverse, converse and contra positive of “ If $\triangle ABC$ is a right angle triangle then $AC^2 = AB^2 + BC^2$ ”	1	2	2
	b).	Translate the following statement into symbolic form. “Any integer is either positive or negative”.	1	2	2
	c).	Define relation matrix and give an example.	2	1	2
	d).	Define Lattice and give an example.	2	1	2
	e).	How many 4 digit numbers divisible by 5 can be formed using the digits 3, 7, 1, 5, 6.	3	2	2
	f).	Solve the recurrence relation $a_n - 5a_{n-1} + 6a_{n-2} = 0, n \geq 2$.	3	3	2
	g).	Define adjacency matrix.	4	1	2
	h).	Define Eulerian graph.	4	1	2
	i).	Define tree and give an example.	5	1	2
	j).	Define Graph coloring.	5	1	2
					5 x 10 =50Marks
		UNIT-1			
2.	a).	Prove that $\{[(p \vee q) \rightarrow r] \wedge \neg p\} \rightarrow (q \rightarrow r)$ is a tautology.	1	3	5
	b).	Verify that the following argument is valid by using the rules of inference If Clifton does not live in France, then he does not speak French. Clifton does not drive a Datsun. If Clifton lives in France, then he rides a bicycle. Either Clifton speaks French, or he drives a Datsun. Hence, Clifton rides a bicycle.	1	3	5
		OR			
3.	a).	Verify that the following argument is valid by using the rules of inference, quantifiers. Babies are illogical. Nobody is despised who can manage a crocodile.	1	3	5

		Illogical people are despised. Hence, babies cannot manage crocodiles.			
	b).	Find the PDNF and PCNF of $p \vee \neg q$	1	3	5
		UNIT-2			
4.	a).	Find the number of integers between 1 and 250 which are divisible by any of the integers 2, 3, 5 or 7.	2	3	5
	b).	Let R denote a relation on the set of ordered pairs of positive integers such that $(x, y)R(u, v)$ if and only if $xv = yu$. Then establish that 'R' is an equivalence relation.	2	3	5
		OR			
5.	a).	Define Hasse diagram and draw Hasse diagram of $(P(A), \subseteq)$, where $A = \{1, 2, 3\}$.	2	3	5
	b).	Define bijective and inverse function. Find the inverse of the function $f(x) = 5x + 2$.	2	3	5
		UNIT-3			
6.	a).	A cricket team of 11 is to be selected out of 14 players of whom 5 are bowlers. Find the number of ways in which this can be done so as to include atleast 3 bowlers.	3	3	5
	b).	i) Determine the term independent of x in the expansion of $\left(x^2 + \frac{1}{x}\right)^{12}$ ii) Determine the coefficient of $x^5 y^{10} z^5 w^5$ in the expansion $(x + 7y + 3z + w)^{25}$	3	3	5
		OR			
7.	a).	How many integral solutions are there to $x_1 + x_2 + x_3 + x_4 + x_5 = 20$ where $x_1 \geq 3, x_2 \geq 2, x_3 \geq 4, x_4 \geq 6, x_5 \geq 0$.	3	3	5
	b).	Solve the recurrence relation $a_n - 5a_{n-1} + 6a_{n-2} = 0, n \geq 2$ using generating functions.	3	3	5
		UNIT-4			
8.		Define isomorphism of graphs and determine the Isomorphism between the following graphs. 	4	3	10
		OR			
9.	a).	Prove that in any graph, there is an even number of vertices of odd degree.	4	3	5
	b).	Define Hamiltonian graph and find the Hamiltonian path and	4	3	5

		Hamiltonian graph in the following graph. 			
		UNIT-5			
10.	a).	State and Prove Euler's formula for planar graphs.	5	3	5
	b).	Show that a tree with n vertices has exactly (n-1) vertices.	5	3	5
		OR			
11.		Find the minimal spanning tree for the following weighted graph 	5	3	10

CO-COURSE OUTCOME

KL-KNOWLEDGE LEVEL

M-MARKS

NOTE: Questions can be given as A,B splits or as a single Question for 10 marks



SRKR
ENGINEERING COLLEGE
AUTONOMOUS

Course Code: B23HS2101					
SAGI RAMA KRISHNAM RAJU ENGINEERING COLLEGE (A)					R23
II B.Tech. I Semester MODEL QUESTION PAPER					
UNIVERSAL HUMAN VALUES-2: UNDERSTANDING HARMONY AND ETHICAL HUMAN CONDUCT					
(Common to all programmes of Engineering)					
Time: 3 Hrs.			Max. Marks: 70 M		
Answer Question No.1 compulsorily					
Answer ONE Question from EACH UNIT					
Assume suitable data if necessary					
10 x 2 = 20 Marks					
			CO	KL	M
1.	a).	What are the basic guidelines for value education?	1	2	2
	b).	What is MBTI personality test?	1	2	2
	c).	How can we differentiate between the needs of the Self and the needs of the Body?	2	2	2
	d).	What are the characteristics and activities of the Self (I)?	2	2	2
	e).	How is 'respect' defined in the context of human interaction?	3	2	2
	f).	How is society described in relation to the family?	3	2	2
	g).	How are the four orders of nature interconnected?	4	2	2
	h).	How does the idea of self-regulation in nature contribute to its harmony?	4	2	2
	i).	Define definitiveness of (ethical) human conduct.	5	2	2
	j).	Explain how humanistic education can influence professional ethics.	5	2	2
5 x 10 = 50 Marks					
		UNIT - I	CO	KL	M
2.	a).	Discuss natural acceptance	1	2	5
	b).	Differentiate prosperity and deprivation	1	2	5
		OR			
3.	a).	Deliberate the right understanding in perspective to self exploration.	1	2	5
	b).	What are the key functions of the MBTI? Explain.	1	2	5
		UNIT - II			
4.	a).	Illustrate coexistence of "I" and "Body ".	1	2	5
	b).	Distinguishing between the Needs of the Self and the Body	1	2	5
		OR			
5.	a).	Discuss Characteristic activities of Harmony with "I".	1	2	5
	b).	Explain Sanyam and Health.	1	2	5

		UNIT - III			
6.	a).	Write a note on human-human relationship as regarding harmony.	2	2	5
	b).	Differentiate intention and competence.	2	2	5
		OR			
7.	a).	Discuss salient values in relationship.	3	2	5
	b).	Illustrate universal Harmonious Society - an Undivided society.	3	2	5
		UNIT - IV			
8.		Discuss orders of life in nature and its significance self regulation of individual	4	2	10
		OR			
9.		Illustrate existence of human being as coexistence with universe in perspective of space	4	2	10
		UNIT - V			
10.		Discuss importance of professional competence for augmenting universal human order.	5	2	10
		OR			
11.	a).	Case study of typical holistic technologies.	5	2	5
	b).	Role of engineer in promoting harmony in society	5	2	5

CO-COURSE OUTCOME

KL-KNOWLEDGE LEVEL

M-MARKS

NOTE: Questions can be given as A,B splits or as a single Question for 10 marks

Course Code: B23AM2101					
SAGI RAMA KRISHNAM RAJU ENGINEERING COLLEGE (A)					R23
II B.Tech. I Semester MODEL QUESTION PAPER					
ARTIFICIAL INTELLIGENCE					
For AIML					
Time: 3 Hrs.			Max. Marks: 70 M		
Answer Question No.1 compulsorily					
Answer ONE Question from EACH UNIT					
Assume suitable data if necessary					
10 x 2 = 20 Marks					
			CO	KL	M
1.	a).	Explain state space representation in the context of AI.	1	2	2
	b).	Name and describe three different domains of AI.	1	2	2
	c).	Explain the differences between informed and uninformed search strategies.	2	2	2
	d).	Explain heuristic search.	2	2	2
	e).	What is prolog?	3	1	2
	f).	What is frame?	3	1	2
	g).	Explain Alpha-Beta pruning.	4	2	2
	h).	What is Well-Formed Formula?	4	1	2
	i).	Explain applications of Expert systems.	5	2	2
	j).	What is Reinforcement Learning?	5	1	2
5 x 10 = 50 Marks					
		UNIT-1	CO	KL	M
2.	a).	Explain about different problem characteristics of AI problem.	1	2	5
	b).	Use state space representation to solve the Water Jug Problem. Describe each step.	1	3	5
		OR			
3.	a).	Explain different types of agents in AI? Provide a brief description of each.	1	2	5
	b).	Apply nearest neighbor heuristic algorithm to solve Travelling Salesmen Problem (TSP).	1	3	5
		UNIT-2			
4.	a).	Explain AO* algorithm with a suitable example.	2	2	5
	b).	Apply simple hill climbing to solve 8-puzzle problem.	2	3	5
		OR			
5.	a).	Differentiate BFS and DFS	2	3	5
	b).	Apply A* algorithm to solve 8-puzzle problem.	2	3	5

		UNIT-3			
6.	a).	Explain Truth maintenance system in detail.	3	2	5
	b).	Represent the following facts in Conceptual Dependency: a. John gave the AI book to marry. b. John punched marry	3	3	5
		OR			
7.	a).	Describe what scripts are in knowledge representation and give an example of a script for a common activity (e.g., going to a restaurant).	3	2	5
	b).	Represent the following facts using Partitioned Semantic Nets: a. The dog bite the mail carrier b. Every batter hit every bowler	3	3	5
		UNIT-4			
8.		Apply the Resolution algorithm to Prove that: ● John likes peanuts. From the following facts: a. John likes all kind of food. b. Apple and vegetable are food. c. Anything anyone eats and not killed is food. d. Anil eats peanuts and still alive. e. Harry eats everything that Anil eats.	4	3	10
		OR			
9.	a).	Apply unification algorithm to the following: a. Like (john, x) Hate (john, x) b. Like (Marcus, Caesar, john) and Like (x, y) c. Like (john, kate) and Like (x, kate)	4	3	5
	b).	Describe the Mini-Max algorithm and its purpose in game playing.	4	2	5
		UNIT-5			
10.	a).	Explain how decision trees are constructed and how they make predictions.	5	2	5
	b).	Explain the architecture of Expert system.	5	2	5
		OR			
11.	a).	Explain the difference between inductive and deductive learning with examples.	5	2	5
	b).	Explain advantages and disadvantages of expert systems.	5	2	5

CO-COURSE OUTCOME

KL-KNOWLEDGE LEVEL

M-MARKS

NOTE: Questions can be given as A,B splits or as a single Question for 10 marks

Course Code: B23CS2102					
SAGI RAMA KRISHNAM RAJU ENGINEERING COLLEGE (A)					R23
II B.Tech. I Semester MODEL QUESTION PAPER					
DATABASE MANAGEMENT SYSTEMS					
Common to CSE, CSG, CSIT & AIML					
Time: 3 Hrs.			Max. Marks: 70 M		
Answer Question No.1 compulsorily					
Answer ONE Question from EACH UNIT					
Assume suitable data if necessary					
10 x 2 = 20 Marks					
			CO	KL	M
1.	a)	Compare DBMS and database.	1	2	2
	b)	Describe briefly two applications of DBMS	1	2	2
	c)	Describe an Entity set in ER model	2	2	2
	d)	Apply relational algebra selection operation for an example query	2	3	2
	e)	Illustrate a nested query in SQL	3	2	2
	f)	Use natural join in SQL to display data in two tables: t1 and t2	3	3	2
	g)	Explain normalization in RDBMS	4	2	2
	h)	Find whether the “E” attribute is key or not for relation R with F = {A->B, B->C, C->D, E->A}?	4	3	2
	i)	Describe Atomicity in brief	5	2	2
	j)	Discuss in brief about the log	5	2	2
5 x 10 = 50 Marks					
		UNIT-1			
2.		Summarize the differences between Database Management System and FPS	1	2	10
		OR			
3.		Illustrate Database system structure	1	2	10
		UNIT-2			
4.		Find different types of keys for the following tables Student(sid, sname, status), Catalog(sid,pid,cost) and Parts (pid, pname, color)	2	3	10
		OR			
5.		Apply conceptual DB design and draw ER diagrams for each situation for the University database containing information about departments (identified by DEPTNO), instructors (identified by SSN) and courses(identified by COURSEID). i. Each department can offer any number of courses	2	3	10

		ii. Many instructors can work in a department iii. An instructor can work only in one department iv. Each instructor can take any number of courses v. A course can be taken by only one instructor			
		UNIT-3			
6.		How do you use the different types of SQL statements (DDL and DML) on tables?	3	3	10
		OR			
7.		Apply different kinds of joins on your own example tables	3	3	10
		UNIT-4			
8.		How do you use functional dependencies to determine whether a table is in 2NF or 3NF?	4	3	10
		OR			
9.		Apply the steps to convert a table into BCNF using one example.	4	3	10
		UNIT-5			
10.		Illustrate the ARIES recovery algorithm with examples.	5	2	10
		OR			
11.		Describe different versions of 2PL protocols	5	2	10

CO-COURSE OUTCOME

KL-KNOWLEDGE LEVEL

M-MARKS

NOTE: Questions can be given as A,B splits or as a single Question for 10 marks

Course Code: B23CS2103					
SAGI RAMA KRISHNAM RAJU ENGINEERING COLLEGE (A)					R23
II B.Tech. I Semester MODEL QUESTION PAPER					
OBJECT ORIENTED PROGRAMMING THROUGH JAVA					
(Common to CSE, CSG, CSIT, AIML & CIC)					
Time: 3 Hrs.			Max. Marks: 70 M		
First Question is compulsory					
Answer ONE Question from EACH UNIT					
Assume suitable data if necessary					
					10 x 2 = 20 Marks
			CO	KL	M
1.	a).	Compare POP and OOP?	1	2	2
	b).	Describe a scenario where using this keyword is necessary in a Java class.	1	2	2
	c).	What is type conversion in Java, and why is it necessary?	2	1	2
	d).	Explain String Constant Pool	2	2	2
	e).	What is Adapter class?	3	1	2
	f).	What are the uses of Super Keyword?	3	1	2
	g).	Explain about throw and throws keyword?	4	2	2
	h).	What are thread priorities in Java, and how do they affect the execution of threads?	4	1	2
	i).	What is File IO in Java?	5	1	2
	j).	List the steps to establish a database connection using JDBC	5	1	2
					5 x 10 = 50 Marks
		UNIT-1	CO	KL	M
2.	a).	Explain Features of JAVA.	1	2	5
	b).	Develop a program to accept two integers as command line arguments and print the sum of the two numbers.	1	3	5
		OR			
3.	a).	Explain Method Overloading with an example.	1	2	5
	b).	Explain different types of Constructors with suitable examples.	1	2	5
		UNIT-2			
4.	a).	Illustrate ArrayList and its methods.	2	2	5
	b).	Develop a program to reverse the elements of a given 2*2 array. Four integer numbers need to be passed as Command Line arguments.	2	3	5
		OR			
5.	a).	Illustrate HashMap and its methods.	2	2	5
	b).	Differentiate String and StringBuffer class.	2	3	5

		UNIT-3			
6.	a).	Create a package called test package. Define a class called foundation inside the test package. Inside the class, you need to define 4 integer variables: var1 with private access modifier, var2 with default access modifier, var3 with protected access modifier and var4 with public access modifier. Import this class and packages in another class. Try to access all 4 variables of the foundation class and see what variables are accessible and what are not accessible.	3	3	5
	b).	Differentiate Interface and Abstract class.	3	3	5
		OR			
7.	a).	Explain different types of inheritance with an example.	3	2	5
	b).	Create an interface Vehicle with a default method message () that returns nothing and prints "Inside Vehicle". Create an interface FourWheeler with a default method message () that returns nothing and prints "Inside FourWheeler". Create a class Car implementing these two interfaces. In this class, Override the message () method and call the message () method of the Vehicle interface using super keyword. Instantiate the class, call the message method and print the output.	3	3	5
		UNIT-4			
8.	a).	Explain Multiple Catch Statements with an example.	4	2	5
	b).	Develop a java program to implement Thread Synchronization.	4	3	5
		OR			
9.	a).	Explain Thread Life Cycle with a neat diagram.	4	2	5
	b).	Develop a program that take an input String from user and parse it to integer, if it is not a number it will throw number format exception Catch it and print "Entered input is not a valid format for an integer." or else print the square of that number.	4	3	5
		UNIT-5			
10.	a).	Develop a program to copy contents of one file to another file.	5	3	5
	b).	Explain different types of JDBC drivers with neat diagrams.	5	2	5
		OR			
11.	a).	Develop a JDBC program to retrieve data from the database.	5	3	5
	b).	Explain the purpose of the Reader and Writer classes in Java. How do they differ from InputStream and OutputStream?	5	2	5

CO-COURSE OUTCOME

KL-KNOWLEDGE LEVEL

M-MARKS

NOTE: Questions can be given as A,B splits or as a single Question for 10 marks

Course Code: B23HS2203					
SAGI RAMA KRISHNAM RAJU ENGINEERING COLLEGE (A)					R23
II B.Tech. II Semester MODEL QUESTION PAPER					
OPTIMIZATION TECHNIQUES					
(Common to AIML & IT)					
Time: 3 Hrs.			Max. Marks: 70 M		
Answer Question No.1 compulsorily					
Answer ONE Question from EACH UNIT					
Assume suitable data if necessary					
10 x 2 = 20 Marks					
			CO	KL	M
1.	a).	State the optimization problem	1	2	2
	b).	Classify various types of optimization problems	1	2	2
	c).	Define feasible region.	2	2	2
	d).	Define artificial variable.	2	2	2
	e).	Name the methods to determine the initial basic feasible solution in TP.	3	2	2
	f).	What is degeneracy in transportation problem?	3	2	2
	g).	Define integer programming problem.	4	2	2
	h).	What are the methods used in solving integer programming problem?	4	2	2
	i).	Define critical activity.	5	2	2
	j).	What are the three main phases of project?	5	2	2
5 x 10 = 50 Marks					
		UNIT-1			
2.		Solve the following nonlinear programming problem, using the Lagrangian multipliers: Optimize $Z=4x_1^2+2x_2^2+x_3^2-4x_1x_2$ Subject to: $x_1+x_2+x_3=15$; $2x_1-x_2+2x_3=20$ and $x_1,x_2,x_3>0$	1	3	10
		OR			
3.		Use the Kuhn-Tucker conditions to solve the following nonlinear programming problem: Maximize $Z=2x_1-x_1^2+x_2$ Subject to the constraints: $2x_1+3x_2\leq 6$ $2x_1+x_2\leq 4$ and $x_1, x_2\geq 0$	1	3	10

		UNIT-2																																	
4.		Solve the following LPP by graphical method Maximize $Z = 8x_1 + 6x_2$ Subject to the constraints: $4x_1 + 2x_2 \leq 60$ $2x_1 + 4x_2 \leq 48$ and $x_1, x_2 \geq 0$	2	3	10																														
		OR																																	
5.		Solve the following LPP by Big-M method Minimize $Z = 4x_1 + 3x_2$ Subject to the constraints: $2x_1 + x_2 \geq 10$ $-3x_1 + 2x_2 \leq 6$ $x_1 + x_2 \geq 6$ and $x_1, x_2 \geq 0$	2	3	10																														
		UNIT-3																																	
6.		Solve the following transportation problem. <table border="1" data-bbox="365 860 1128 1095"> <tr> <td>Destination → Origin ↓</td><td>D₁</td><td>D₂</td><td>D₃</td><td>D₄</td><td>Supply</td></tr> <tr> <td>O₁</td><td>5</td><td>3</td><td>6</td><td>2</td><td>19</td></tr> <tr> <td>O₂</td><td>4</td><td>7</td><td>9</td><td>1</td><td>37</td></tr> <tr> <td>O₃</td><td>3</td><td>4</td><td>7</td><td>5</td><td>34</td></tr> <tr> <td>Demand</td><td>16</td><td>18</td><td>31</td><td>25</td><td>90</td></tr> </table>	Destination → Origin ↓	D ₁	D ₂	D ₃	D ₄	Supply	O ₁	5	3	6	2	19	O ₂	4	7	9	1	37	O ₃	3	4	7	5	34	Demand	16	18	31	25	90	3	3	10
Destination → Origin ↓	D ₁	D ₂	D ₃	D ₄	Supply																														
O ₁	5	3	6	2	19																														
O ₂	4	7	9	1	37																														
O ₃	3	4	7	5	34																														
Demand	16	18	31	25	90																														
		OR																																	
7.		Find the initial solution by VAM and optimality by MODI method for the following transportation problem. <table border="1" data-bbox="392 1261 1094 1561"> <tr> <td>Warehouse → Factory ↓</td><td>W₁</td><td>W₂</td><td>W₃</td><td>W₄</td><td>Capacity (Supply)</td></tr> <tr> <td>F₁</td><td>21</td><td>16</td><td>25</td><td>13</td><td>11</td></tr> <tr> <td>F₂</td><td>17</td><td>18</td><td>14</td><td>23</td><td>13</td></tr> <tr> <td>F₃</td><td>32</td><td>27</td><td>18</td><td>41</td><td>19</td></tr> <tr> <td>Requirement (Demand)</td><td>6</td><td>10</td><td>12</td><td>15</td><td>43</td></tr> </table>	Warehouse → Factory ↓	W ₁	W ₂	W ₃	W ₄	Capacity (Supply)	F ₁	21	16	25	13	11	F ₂	17	18	14	23	13	F ₃	32	27	18	41	19	Requirement (Demand)	6	10	12	15	43	3	3	10
Warehouse → Factory ↓	W ₁	W ₂	W ₃	W ₄	Capacity (Supply)																														
F ₁	21	16	25	13	11																														
F ₂	17	18	14	23	13																														
F ₃	32	27	18	41	19																														
Requirement (Demand)	6	10	12	15	43																														
		UNIT-4																																	
8.		Use Branch and Bound algorithm to solve the following problem. Maximize $Z = 7x_1 + 9x_2$ Subject to the constraints: $-x_1 + 3x_2 \leq 6$ $7x_1 + x_2 \leq 35$ and $x_1, x_2 \geq 0$ and are integers.	4	3	10																														
		OR																																	
9.		Use Gomory's cutting plane algorithm to solve the following problem.	4	3	10																														

		Maximize $Z= x_1 + x_2$ Subject to the constraints: $7x_1 -5x_2 \leq 7$ $-12x_1 + 15x_2 \leq 7$ and $x_1, x_2 \geq 0$ and are integers.																																											
		UNIT-5																																											
10.		The following are the time estimates and the precedence relationships of the activities in a project network: <table><tr><td>Activity</td><td>A</td><td>B</td><td>C</td><td>D</td><td>E</td><td>F</td><td>G</td><td>H</td><td>I</td><td>J</td><td>K</td></tr><tr><td>Predecessor activity</td><td>-</td><td>-</td><td>-</td><td>A</td><td>B</td><td>B</td><td>C</td><td>E</td><td>D</td><td>F,G</td><td>H,I</td></tr><tr><td>Timeestimate (weeks)</td><td>4</td><td>7</td><td>3</td><td>6</td><td>4</td><td>7</td><td>6</td><td>10</td><td>3</td><td>4</td><td>2</td></tr></table> Draw the project network diagram. Determine the critical path and the project completion time.	Activity	A	B	C	D	E	F	G	H	I	J	K	Predecessor activity	-	-	-	A	B	B	C	E	D	F,G	H,I	Timeestimate (weeks)	4	7	3	6	4	7	6	10	3	4	2	5	3	10				
Activity	A	B	C	D	E	F	G	H	I	J	K																																		
Predecessor activity	-	-	-	A	B	B	C	E	D	F,G	H,I																																		
Timeestimate (weeks)	4	7	3	6	4	7	6	10	3	4	2																																		
		OR																																											
11.		The time estimates (in weeks) and other characteristics of a project are given below. <table><tr><td>Activity</td><td>1-2</td><td>1-6</td><td>2-3</td><td>2-4</td><td>3-5</td><td>4-5</td><td>6-7</td><td>5-8</td><td>7-8</td></tr><tr><td>Optimistic time</td><td>3</td><td>2</td><td>6</td><td>4</td><td>8</td><td>3</td><td>3</td><td>2</td><td>8</td></tr><tr><td>Most likely time</td><td>6</td><td>5</td><td>12</td><td>5</td><td>11</td><td>7</td><td>9</td><td>4</td><td>16</td></tr><tr><td>Pessimistic time</td><td>9</td><td>8</td><td>18</td><td>6</td><td>14</td><td>11</td><td>15</td><td>6</td><td>18</td></tr></table> i) Find expected activity times and their variances. ii) Calculate expected time and variance of the critical path.	Activity	1-2	1-6	2-3	2-4	3-5	4-5	6-7	5-8	7-8	Optimistic time	3	2	6	4	8	3	3	2	8	Most likely time	6	5	12	5	11	7	9	4	16	Pessimistic time	9	8	18	6	14	11	15	6	18	5	3	10
Activity	1-2	1-6	2-3	2-4	3-5	4-5	6-7	5-8	7-8																																				
Optimistic time	3	2	6	4	8	3	3	2	8																																				
Most likely time	6	5	12	5	11	7	9	4	16																																				
Pessimistic time	9	8	18	6	14	11	15	6	18																																				

CO-COURSE OUTCOME

KL-KNOWLEDGE LEVEL

M-MARKS

NOTE: Questions can be given as A,B splits or as a single Question for 10 marks

Course Code: B23BS2201										
SAGI RAMA KRISHNAM RAJU ENGINEERING COLLEGE (A)										R23
II B.Tech. II SEMESTER MODEL QUESTION PAPER										
PROBABILITY AND STATISTICS										
Common For AIML, CSE, CSBS, CSIT & IT										
Time: 3 Hrs.						Max. Marks: 70 M				
Answer Question No.1 compulsorily										
Answer ONE Question from EACH UNIT										
Assume suitable data if necessary										
10 x 2 = 20Marks										
				</						

	b)	Let X be a random variable with the following probability distribution <table><tr><td>X</td><td>6</td><td>9</td><td>12</td></tr><tr><td>P(X=x)</td><td>1/6</td><td>1/2</td><td>1/3</td></tr></table> Find E(x) and V(x) using the laws of expectations.	X	6	9	12	P(X=x)	1/6	1/2	1/3	2	3	5														
X	6	9	12																								
P(X=x)	1/6	1/2	1/3																								
		OR																									
5.	a)	A manufacturer of blades knows that 5% of his product is defective. If he sells blades in boxes of 100 and guarantees that not more than 10 blades will be defective. Find the probability that a box will fail to meet the guaranteed quality?	2	3	5																						
	b)	In a distribution exactly normal, 11.03% of the items are under 25-kilogram weight and 89.97% of the items are under 70-kilogram weight. What are the mean and standard deviation the distribution.	2	3	5																						
		UNIT-3																									
6.	a)	If the regression lines are $2y - x = 50$ and $3y - 2x = 10$, then find the mean values of x, y and the correlation coefficient between x and y.	3	3	5																						
	b)	The marks obtained by 10 students in Mathematics(X) and Statistics(Y) are given below. <table><tr><td>X</td><td>68</td><td>64</td><td>75</td><td>50</td><td>64</td><td>80</td><td>75</td><td>40</td><td>55</td><td>64</td></tr><tr><td>Y</td><td>62</td><td>58</td><td>68</td><td>45</td><td>81</td><td>60</td><td>68</td><td>48</td><td>50</td><td>70</td></tr></table> Find the rank correlation coefficient between the subjects.	X	68	64	75	50	64	80	75	40	55	64	Y	62	58	68	45	81	60	68	48	50	70	3	3	5
X	68	64	75	50	64	80	75	40	55	64																	
Y	62	58	68	45	81	60	68	48	50	70																	
		OR																									
7.		Obtain a second-degree polynomial from the following data. <table><tr><td>X</td><td>0</td><td>1</td><td>2</td><td>3</td><td>4</td><td>5</td><td>6</td><td>7</td><td>8</td></tr><tr><td>Y</td><td>12</td><td>10.5</td><td>10</td><td>8</td><td>7</td><td>8</td><td>7.5</td><td>8.5</td><td>9</td></tr></table>	X	0	1	2	3	4	5	6	7	8	Y	12	10.5	10	8	7	8	7.5	8.5	9	3	3	10		
X	0	1	2	3	4	5	6	7	8																		
Y	12	10.5	10	8	7	8	7.5	8.5	9																		
		UNIT-4																									
8.	a)	A random sample of 10 ball bearings produced by a company have a mean diameter of 0.5060 cm with s.d. 0.004 cm. Find the maximum error estimate E and 95% confidence interval for the actual mean of ball bearings produced by this company assuming sampling from normal population.	4	3	5																						
	b)	Determine a 95% confidence interval for the variance of a normal population with the sample: 145.3, 145.1, 145.4 and 146.2.	4	3	5																						
		OR																									
9.		A population consists of five numbers 2, 3, 6, 8 and 11, Consider all possible samples of size two which can be drawn with replacement for the population, Calculate (a) The mean of the population (b) The standard deviation of the population (c) The mean of the sampling distribution of means (d) The standard deviation of the sampling distribution of means.	4	3	10																						

		UNIT-5																			
10.	a)	In a large city A, 20 percent of a random sample of 900 school children had defective eye-sight. In other large city B, 15 percent of random sample of 1600 children had the same defect. Is this difference between the two proportions significant?	5	3	5																
	b)	A group of 10 boys fed on a diet A and another group of 8 boys fed on a different diet B recorded the following increase in weights. Diet A (in Kg.) : 5 6 8 1 12 4 3 9 6 10 Diet B (in Kg.): 2 3 6 8 10 1 2 8 Test whether the variance of weights differ significantly?	5	3	5																
		OR																			
11.		Fit a Poisson distribution to the given data and test the goodness of fit. <table><tr><td>x</td><td>0</td><td>1</td><td>2</td><td>3</td><td>4</td><td>5</td><td>6</td></tr><tr><td>f</td><td>275</td><td>72</td><td>30</td><td>7</td><td>5</td><td>2</td><td>1</td></tr></table>	x	0	1	2	3	4	5	6	f	275	72	30	7	5	2	1	5	3	10
x	0	1	2	3	4	5	6														
f	275	72	30	7	5	2	1														

CO-COURSE OUTCOME

KL-KNOWLEDGE LEVEL

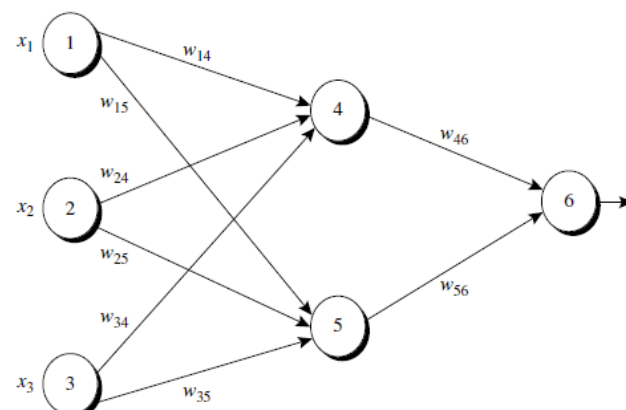
M-MARKS

NOTE: Questions can be given as A,B splits or as a single Question for 10 marks



Course Code:B23AM2201																				
SAGI RAMA KRISHNAM RAJU ENGINEERING COLLEGE (A)					R23															
II B.Tech. II Semester MODEL QUESTION PAPER																				
MACHINE LEARNING																				
For AIML																				
Time: 3 Hrs.			Max. Marks: 70 M																	
Answer Question No.1 compulsorily																				
Answer ONE Question from EACH UNIT																				
Assume suitable data if necessary																				
10 x 2 = 20 Marks																				
			CO	KL	M															
1.	a)	Define Reinforcement Learning.	1	1	2															
	b)	Explain about Eigen values using example.	1	2	2															
	c)	Illustrate about various proximity measures.	2	2	2															
	d)	Define Regression.	2	1	2															
	e)	Explain about Bias Variance Trade-off.	3	2	2															
	f)	Define Multi-Class Classification.	3	1	2															
	g)	Illustrate various linear discriminant models.	4	2	2															
	h)	Define Perceptron.	4	1	2															
	i)	Apply Euclidean distance in clustering for points A: (1, 2), B: (4,6)	5	3	2															
	j)	Define Divisive Clustering.	5	1	2															
5 x 10 =50Marks																				
UNIT-1																				
2.	a)	Explain the stages in Machine Learning	1	2	5															
	b)	Illustrate the types of data using examples	1	3	5															
OR																				
3.	a)	Explain filter method in feature selection	1	2	5															
	b)	Apply PCA on the following dataset	1	3	5															
		<table><tr><td>Feature</td><td>Example1</td><td>Example2</td><td>Example3</td><td>Example4</td></tr><tr><td>A</td><td>4</td><td>8</td><td>13</td><td>7</td></tr><tr><td>B</td><td>11</td><td>4</td><td>5</td><td>14</td></tr></table>				Feature	Example1	Example2	Example3	Example4	A	4	8	13	7	B	11	4	5	14
Feature		Example1				Example2	Example3	Example4												
A		4				8	13	7												
B	11	4	5	14																
UNIT-2																				
4.	a)	Explain about Non-Metric Similarity Functions.	2	2	5															
	b)	Apply K-Nearest Neighbor Classifier to following dataset.	2	3	5															

		<table><tr><th>BRIGHTNESS</th><th>SATURATION</th><th>CLASS</th></tr><tr><td>40</td><td>20</td><td>Red</td></tr><tr><td>50</td><td>50</td><td>Blue</td></tr><tr><td>60</td><td>90</td><td>Blue</td></tr><tr><td>10</td><td>25</td><td>Red</td></tr><tr><td>70</td><td>70</td><td>Blue</td></tr><tr><td>60</td><td>10</td><td>Red</td></tr><tr><td>25</td><td>80</td><td>Blue</td></tr></table> Now identify to which class label the given test sample belongs to considering K=3 nearest neighbors (Brightness=20, Saturation=35, Class=?).	BRIGHTNESS	SATURATION	CLASS	40	20	Red	50	50	Blue	60	90	Blue	10	25	Red	70	70	Blue	60	10	Red	25	80	Blue																																																																					
BRIGHTNESS	SATURATION	CLASS																																																																																													
40	20	Red																																																																																													
50	50	Blue																																																																																													
60	90	Blue																																																																																													
10	25	Red																																																																																													
70	70	Blue																																																																																													
60	10	Red																																																																																													
25	80	Blue																																																																																													
		OR																																																																																													
5.	a)	Explain Radius Distance Nearest Neighbor Algorithm with an example	2	3	5																																																																																										
	b)	Illustrate KNN Regression with an example	2	2	5																																																																																										
		UNIT-3																																																																																													
6.	a)	Apply Information gain metric to generate decision tree for the following dataset <table><tr><th>RID</th><th>age</th><th>income</th><th>student</th><th>credit_rating</th><th>Class: buys_computer</th></tr><tr><td>1</td><td>youth</td><td>high</td><td>no</td><td>fair</td><td>no</td></tr><tr><td>2</td><td>youth</td><td>high</td><td>no</td><td>excellent</td><td>no</td></tr><tr><td>3</td><td>middle_aged</td><td>high</td><td>no</td><td>fair</td><td>yes</td></tr><tr><td>4</td><td>senior</td><td>medium</td><td>no</td><td>fair</td><td>yes</td></tr><tr><td>5</td><td>senior</td><td>low</td><td>yes</td><td>fair</td><td>yes</td></tr><tr><td>6</td><td>senior</td><td>low</td><td>yes</td><td>excellent</td><td>no</td></tr><tr><td>7</td><td>middle_aged</td><td>low</td><td>yes</td><td>excellent</td><td>yes</td></tr><tr><td>8</td><td>youth</td><td>medium</td><td>no</td><td>fair</td><td>no</td></tr><tr><td>9</td><td>youth</td><td>low</td><td>yes</td><td>fair</td><td>yes</td></tr><tr><td>10</td><td>senior</td><td>medium</td><td>yes</td><td>fair</td><td>yes</td></tr><tr><td>11</td><td>youth</td><td>medium</td><td>yes</td><td>excellent</td><td>yes</td></tr><tr><td>12</td><td>middle_aged</td><td>medium</td><td>no</td><td>excellent</td><td>yes</td></tr><tr><td>13</td><td>middle_aged</td><td>high</td><td>yes</td><td>fair</td><td>yes</td></tr><tr><td>14</td><td>senior</td><td>medium</td><td>no</td><td>excellent</td><td>no</td></tr></table>	RID	age	income	student	credit_rating	Class: buys_computer	1	youth	high	no	fair	no	2	youth	high	no	excellent	no	3	middle_aged	high	no	fair	yes	4	senior	medium	no	fair	yes	5	senior	low	yes	fair	yes	6	senior	low	yes	excellent	no	7	middle_aged	low	yes	excellent	yes	8	youth	medium	no	fair	no	9	youth	low	yes	fair	yes	10	senior	medium	yes	fair	yes	11	youth	medium	yes	excellent	yes	12	middle_aged	medium	no	excellent	yes	13	middle_aged	high	yes	fair	yes	14	senior	medium	no	excellent	no	3	3	10
RID	age	income	student	credit_rating	Class: buys_computer																																																																																										
1	youth	high	no	fair	no																																																																																										
2	youth	high	no	excellent	no																																																																																										
3	middle_aged	high	no	fair	yes																																																																																										
4	senior	medium	no	fair	yes																																																																																										
5	senior	low	yes	fair	yes																																																																																										
6	senior	low	yes	excellent	no																																																																																										
7	middle_aged	low	yes	excellent	yes																																																																																										
8	youth	medium	no	fair	no																																																																																										
9	youth	low	yes	fair	yes																																																																																										
10	senior	medium	yes	fair	yes																																																																																										
11	youth	medium	yes	excellent	yes																																																																																										
12	middle_aged	medium	no	excellent	yes																																																																																										
13	middle_aged	high	yes	fair	yes																																																																																										
14	senior	medium	no	excellent	no																																																																																										
		OR																																																																																													
7.	a)	Consider the following training data and classify the pattern using Naive Bayes Classifier. <table><tr><th>Pattern</th><th>Feature1</th><th>Feature2</th><th>Feature3</th><th>Class</th></tr><tr><td>1</td><td>0</td><td>0</td><td>0</td><td>C0</td></tr><tr><td>2</td><td>1</td><td>0</td><td>1</td><td>C1</td></tr><tr><td>3</td><td>1</td><td>0</td><td>0</td><td>C0</td></tr><tr><td>4</td><td>1</td><td>1</td><td>1</td><td>C1</td></tr><tr><td>5</td><td>0</td><td>1</td><td>1</td><td>C1</td></tr><tr><td>6</td><td>0</td><td>1</td><td>1</td><td>C0</td></tr><tr><td>7</td><td>1</td><td>0</td><td>1</td><td>?</td></tr></table>	Pattern	Feature1	Feature2	Feature3	Class	1	0	0	0	C0	2	1	0	1	C1	3	1	0	0	C0	4	1	1	1	C1	5	0	1	1	C1	6	0	1	1	C0	7	1	0	1	?	3	3	10																																																		
Pattern	Feature1	Feature2	Feature3	Class																																																																																											
1	0	0	0	C0																																																																																											
2	1	0	1	C1																																																																																											
3	1	0	0	C0																																																																																											
4	1	1	1	C1																																																																																											
5	0	1	1	C1																																																																																											
6	0	1	1	C0																																																																																											
7	1	0	1	?																																																																																											

		UNIT-4																																	
8.	a)	Explain Perceptron Learning algorithm	4	2	5																														
	b)	Illustrate Support Vector Machines with kernel trick.	4	2	5																														
		OR																																	
9.	a)	Explain about Linear Regression	4	2	5																														
	b)	Apply backpropagation algorithm on following MLP and data. Show the weights values after single epoch. <table><tr><th>x_1</th><th>x_2</th><th>x_3</th><th>w_{14}</th><th>w_{15}</th><th>w_{24}</th><th>w_{25}</th><th>w_{34}</th><th>w_{35}</th><th>w_{46}</th><th>w_{56}</th><th>θ_4</th><th>θ_5</th><th>θ_6</th></tr><tr><td>1</td><td>0</td><td>1</td><td>0.2</td><td>-0.3</td><td>0.4</td><td>0.1</td><td>-0.5</td><td>0.2</td><td>-0.3</td><td>-0.2</td><td>-0.4</td><td>0.2</td><td>0.1</td></tr></table>	x_1	x_2	x_3	w_{14}	w_{15}	w_{24}	w_{25}	w_{34}	w_{35}	w_{46}	w_{56}	θ_4	θ_5	θ_6	1	0	1	0.2	-0.3	0.4	0.1	-0.5	0.2	-0.3	-0.2	-0.4	0.2	0.1	4	3	5		
x_1	x_2	x_3	w_{14}	w_{15}	w_{24}	w_{25}	w_{34}	w_{35}	w_{46}	w_{56}	θ_4	θ_5	θ_6																						
1	0	1	0.2	-0.3	0.4	0.1	-0.5	0.2	-0.3	-0.2	-0.4	0.2	0.1																						
		UNIT-5																																	
10.	a)	Explain Agglomerative Clustering method	5	2	5																														
	b)	Apply K-Means Clustering with two iterations to form clusters by taking the initial cluster centers as points P1 and P4 on the following dataset to partition into two clusters. P1(1,1), P2(2,2), P3(3,4), P4(5,7), P5(3,5), P6(4,5), P7(4,6).	5	3	5																														
		OR																																	
11.	a)	Apply Fuzzy C-Means Clustering on the following dataset. Let number of clusters $C=2$ and $M=2$. <table><tr><td></td><td>X1</td><td>X2</td><td>X3</td><td>X4</td></tr><tr><td>C1</td><td>1</td><td>1</td><td>0</td><td>0</td></tr><tr><td>C2</td><td>0</td><td>0</td><td>1</td><td>1</td></tr></table> <table><tr><td>Datapoints</td><td>F1</td><td>F2</td></tr><tr><td>X1</td><td>1</td><td>1</td></tr><tr><td>X2</td><td>2</td><td>2</td></tr><tr><td>X3</td><td>4</td><td>3</td></tr><tr><td>X4</td><td>5</td><td>3</td></tr></table>		X1	X2	X3	X4	C1	1	1	0	0	C2	0	0	1	1	Datapoints	F1	F2	X1	1	1	X2	2	2	X3	4	3	X4	5	3	5	3	5
	X1	X2	X3	X4																															
C1	1	1	0	0																															
C2	0	0	1	1																															
Datapoints	F1	F2																																	
X1	1	1																																	
X2	2	2																																	
X3	4	3																																	
X4	5	3																																	
	b)	Illustrate Spectral Clustering method	5	2	5																														

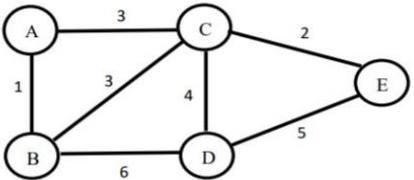
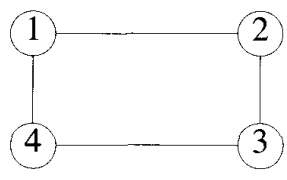
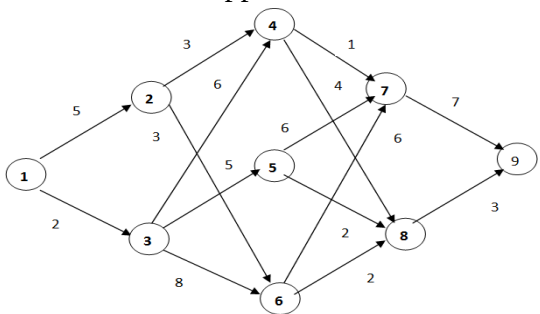
CO-COURSE OUTCOME

KL-KNOWLEDGE LEVEL

M-MARKS

NOTE: Questions can be given as A,B splits or as a single Question for 10 marks

Course Code: B23CS2202					
SAGI RAMA KRISHNAM RAJU ENGINEERING COLLEGE (A)					R23
II B.Tech. II Semester MODEL QUESTION PAPER					
ADVANCED DATA STRUCTURES AND ALGORITHM ANALYSIS					
Common to CSE, CSIT, AIML & CSG					
Time: 3 Hrs.			Max. Marks: 70 M		
Answer Question No.1 compulsorily					
Answer ONE Question from EACH UNIT					
Assume suitable data if necessary					
10 x 2 = 20 Marks					
			CO	KL	M
1.	a).	List out the properties of a Red-Black Tree	1	2	2
	b).	How to check Bi-connectivity in a graph?	1	2	2
	c).	Describe Asymptotic notations.	2	2	2
	d).	Explain the general method of divide and conquer.	2	2	2
	e).	How does the Greedy Method solve the Fractional Knapsack Problem?	3	2	2
	f).	Outline Backtracking strategy.	3	2	2
	g).	Why is Dynamic Programming more efficient than simple recursion for certain problems?	4	2	2
	h).	Explain the role of state and transition in dynamic programming with an example.	4	2	2
	i).	Discuss NP-Hard problems.	5	2	2
	j).	Describe Chromatic Number Decision Problem (CNDP).	5	2	2
5 x 10 = 50 Marks					
		UNIT-1			
2.		Explain AVL Tree rotations with examples	1	2	10
		OR			
3.	a).	Compare and contrast adjacency matrix and adjacency list representations of graphs.	1	2	5
	b).	Describe the Breadth-First Search (BFS) algorithm.	1	2	5
		UNIT-2			
4.		Describe the Quick Sort algorithm and analyze its average-case time complexity.	2	4	10
		OR			
5.		Explain the Divide and Conquer approach for finding the Convex Hull of a set of points and analyze its time complexity.	2	4	10

		UNIT-3			
6.	a).	Solve the following job sequencing with deadlines using greedy method. $n=6, (p_1, p_2, \dots, p_6) = (20, 40, 5, 15, 10, 8)$ and $(d_1, d_2, \dots, d_6) = (5, 2, 4, 3, 3, 1)$	3	3	5
	b).	Find the Minimum Spanning for the following graph using Kruskal's algorithm. 	3	3	5
		OR			
7.	a).	Explain Graph Coloring problem and Construct State Space Tree when $n=4$ and $m=3$. 	3	3	5
	b).	Solve 0/1 knapsack problem for the following data using Back Tracking. $n=5, (p_1, p_2, p_3, p_4, p_5) = (7, 8, 9, 11, 12), (w_1, w_2, w_3, w_4, w_5) = (13, 15, 16, 23, 24)$ with knapsack capacity, $m=26$.	3	3	5
		UNIT-4			
8.		Build OBST for given data $n=4, (a_1, a_2, a_3, a_4) = (\text{do}, \text{if}, \text{int}, \text{while})$ and $p(1:4) = (3, 3, 1, 1)$ $q(0:4) = (2, 3, 1, 1, 1)$.	4	3	10
		OR			
9.		Find a minimum cost path from S to T graph of following figure. Do this in forward Approach. 	4	3	10
		UNIT-5			
10.	a).	Consider the travelling sales person instance defined by the cost matrix. Find out minimum cost path using LC Branch-Bound.	5	3	5

		$\begin{bmatrix} \infty & 2 & 10 & 5 \\ 2 & \infty & 9 & \infty \\ 4 & 3 & \infty & 4 \\ 6 & 8 & 7 & \infty \end{bmatrix}$			
	b).	State and explain cook's theorem	5	2	5
		OR			
11.	a).	Draw the portion of the State Space Tree generated by LCBB for the instances: $n=5, m=12, (p1 \dots p5)=(10,15,6,8,4), (w1 \dots w5)=(4,6,3,4,2)$	5	3	5
	b).	Explain Classes NP-hard and NP-complete	5	2	5
CO-COURSE OUTCOME			KL-KNOWLEDGE LEVEL		M-MARKS

NOTE: Questions can be given as A,B splits or as a single Question for 10 marks



SRKR
ENGINEERING COLLEGE
AUTONOMOUS

Course Code:B23AM2202					
SAGI RAMA KRISHNAM RAJU ENGINEERING COLLEGE (A)					R23
II B.Tech. II Semester MODEL QUESTION PAPER					
DIGITAL LOGIC & COMPUTER ORGANIZATION					
For AIML					
Time: 3 Hrs.			Max. Marks: 70 M		
Answer Question No.1 compulsorily					
Answer ONE Question from EACH UNIT					
Assume suitable data if necessary					
					10 x 2 = 20 Marks
			CO	KL	M
1.	a).	Solve the following Boolean function to a minimum number of literals: $xy + x'z + yz$	1	3	2
	b).	Develop a 2X4 line Decoder.	1	3	2
	c).	Explain various types of buses?	2	2	2
	d).	Differentiate multi computers and multi processors.	2	2	2
	e).	Explain Instruction Format.	3	2	2
	f).	Describe zero address instruction Format with example.	3	2	2
	g).	Explain memory hierarchy.	4	2	2
	h).	Compare static RAM and dynamic RAM	4	2	2
	i).	Explain the need of I/O interface module.	5	2	2
	j).	Summarize various types of output peripheral devices.	5	2	2
5 x 10 = 50 Marks					
		UNIT-1			
2.		Solve the Boolean Function $F(w, x, y, z) = \sum (1, 3, 7, 11, 15)$ into SOP and POS using K-Map.	1	3	10
		OR			
3.		Develop a Combinational Circuit for 8X1 line Multiplexer.	1	3	10
		UNIT-2			
4.		Construct an Arithmetic Logic Shift Unit.	2	3	10
		OR			
5.		Determine Booth Multiplication algorithm with example.	2	3	5
		UNIT-3			
6.	a).	Explain the organization of registers.	3	2	5
	b).	Describe about the Address Sequencing.	3	2	5
		OR			
7.		Illustrate different addressing modes with example.	3	2	10

